Flow Sorting and Sequencing Meadow Fescue Chromosome 4F[1][C][W]

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The analysis of large genomes is hampered by a high proportion of repetitive DNA, which makes the assembly of short sequence reads difficult. This is also the case in meadow fescue (*Festuca pratensis*), which is known for good abiotic stress resistance and has been used in intergeneric hybridization with ryegrasses (*Lolium* spp.) to produce Festulolium cultivars. In this work, we describe a new approach to analyze the large genome of meadow fescue, which involves the reduction of sample complexity without compromising information content. This is achieved by dissecting the genome to smaller parts: individual chromosomes and groups of chromosomes. As the first step, we flow sorted chromosome 4F and sequenced it by Illumina with approximately 50× coverage. This provided, to our knowledge, the first insight into the composition of the fescue genome, enabled the construction of the virtual gene order of the chromosome, and facilitated detailed comparative analysis with the sequenced genomes of rice (*Oryza sativa*), *Brachypodium distachyon*, sorghum (*Sorghum bicolor*), and barley (*Hordeum vulgare*). Using GenomeZipper, we were able to confirm the collinearity of chromosome 4F with barley chromosome 4H and the long arm of chromosome 5H. Several new tandem repeats were identified and physically mapped using fluorescence in situ hybridization. They were found as robust cytogenetic markers for karyotyping of meadow fescue and ryegrass species and their hybrids. The ability to purify chromosome 4F opens the way for more efficient analysis of genomic loci on this chromosome underlying important traits, including freezing tolerance. Our results confirm that next-generation sequencing of flow-sorted chromosomes enables an overview of chromosome structure and evolution at a resolution never achieved before.

Meadow fescue (*Festuca pratensis*) is a grass typically found in hygrophylic and mesophylic meadows and pastures. It provides high-quality forage in pure stands and in grass-clover mixtures. Besides its usage on temporary meadows and pastures, this species is preferentially used as a component of permanent grasslands, where it produces forage during the first 3 years. Its outstanding winter survival is why this species is the major component of grasslands in northern Europe and Canada, and for the same reason, it is used widely in mountain regions of central and western Europe (Germany, Austria, Switzerland, Czech Republic, and Slovakia). However, meadow fescue is less adapted and competitive in lowland coastal regions in the temperate zone. In such climates, it is being replaced by interspecific hybrids of meadow fescue (or tall fescue) with ryegrasses (*Lolium* spp.), especially with *Lolium multiflorum*.

Despite the agronomic importance of meadow fescue, either as a stand-alone crop or as a parent in intergeneric hybridization, the progress in genetics and genomics of meadow fescue is lagging far behind most other crops. Genomic studies in meadow fescue are complicated by the large genome size (3,175 Mb/1C; Kopecký et al., 2010) and a rather symmetrical karyotype. Relative to cereals, and even to closely related ryegrasses, genetic and genomic resources of meadow fescue are limited.

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The only existing genetic map was developed by Alm et al. (2003) and consists of 466 RFLP, amplified fragment length polymorphism (AFLP), and isozyme markers with a total length of 658.8 centimorgan (cM). This map was recently enriched by 149 Diversity Arrays Technology (DArT) markers (Bartoš et al., 2011). So far, two bacterial artificial chromosome (BAC) libraries have been developed for meadow fescue. Donnison et al. (2005) constructed a five-genome coverage library and used it to clone candidate orthologous sequences to the CONSTANS-like rice (*Oryza sativa*) Heading date1 (*Hd1*; Photosensitivity1 [*Se1*]) gene in *Lolium perenne* and meadow fescue, while Kopecký et al. (2010) used a partial genomic BAC library (about one genome coverage) as a source of cytogenetic markers. The only existing DNA array was developed by Kopecký et al. (2009) using DArT technology (Jaccoud et al., 2001), and it consists of 7,680 probes derived from methyl-filtered genomic representations of five species from the *Festuca-Lolium* species complex (*L. perenne*, *L. multiflorum*, *F. pratensis*, *Festuca arundinacea*, and *Festuca glaucescens*). It has been used for diversity studies (Kopecký et al., 2009, 2011; Baird et al., 2012), genetic mapping (Bartoš et al., 2011; Tomaszewski et al., 2012), analysis of genomic composition of Festulolium cultivars (Kopecký et al., 2011), and association analysis of agronomic traits (Bartoš et al., 2011). Sequencing 620 genetically mapped DArT markers provided the first large source of DNA sequences for meadow fescue. However, DArT markers are believed to originate primarily from genic regions due to a methyl-filtration step in the preparation of genomic representations. This was proved by BLAST analysis against EST and hypothetical protein databases (Bartoš et al., 2011). Thus, the DArT markers did not provide a complete insight into the genome composition. Recently, 454 transcriptome sequencing of two mapping parents of meadow fescue has been reported by Vigeland et al. (2013).

Recent advances in next-generation sequencing (NGS) technologies, which enable the sequencing of entire genomes, make it possible to analyze genome structure at high resolution and perform detailed comparative studies across species. NGS data have been widely used for molecular marker development, phylogenetic and ecological studies, and analysis of transcriptomes using RNA sequencing (Egan et al., 2012). Unfortunately, the NGS of whole genomes has limited use in comparative genomic studies in species with no or poor genetic maps and in species for which a reference genome sequence is not available. In such cases, sequencing flow-sorted chromosomes was found to be an extremely valuable approach. For example, NGS of sorted chromosomes has been used to study structural genome changes in cereals during their evolution and speciation. Mayer et al. (2009, 2011) used a novel approach called GenomeZipper to deduce the putative gene order in barley (*Hordeum vulgare*). This approach is based on the comparisons of chromosomal shotgun sequences against model genomes (rice, sorghum [*Sorghum bicolor*], and *Brachypodium distachyon*) to detect syntenic regions. Similarly, Vitulo et al. (2011), Wicker et al. (2011), and Hernandez et al. (2012) used GenomeZipper for comparative studies of wheat (*Triticum aestivum*) homologous group 1 and wheat chromosomes 5A and 4A with rice, sorghum, *B. distachyon*, and a virtual barley genome. This analysis revealed considerable restructuring of wheat chromosome 4A and confirmed the utility of this approach.

Besides the use of flow-sorted chromosomes for NGS, they can be used for several other applications. A number of chromosome-specific BAC libraries were constructed for hexaploid wheat and rye (*Secale cereale*; Šafář et al., 2010), and many of these libraries are used for physical mapping (Paux et al., 2008). Moreover, sorted chromosomes are valuable templates for cytogenetic mapping using fluorescence in situ hybridization (FISH; Kubaláková et al., 2003; Suchánková et al., 2006). Sorted chromosomes were also used as an efficient source of genetic markers for the saturation of genetic maps (Wenzl et al., 2010).

In this study, we set out to explore the possibilities of applying chromosome genomics in meadow fescue. We have developed flow cytogenetics in this species and used flow cytometric sorting to dissect the meadow fescue genome into individual chromosomes. NGS of isolated chromosomes provided, to our knowledge, the first insights into the genome structure of this species and permitted the analysis of collinearity at a high resolution level.

RESULTS

Cell Cycle Synchronization and Preparation of Chromosome Suspension

The experiments performed to optimize the induction of cell cycle synchrony and the accumulation of metaphases in meristem root tips were based on our previous experience with cereals and legumes. The optimal concentration of hydroxyurea (HU) was 1.5 mM; lower concentration (1 mM) did not completely block the cycle. On the other hand, the cell cycle did not recover after blockage by 2 mM HU. The highest metaphase index (55%) was reached 5 h after removal from the DNA synthesis inhibitor (recovery time).

Chromosome Analysis and Sorting

Flow cytometry of mitotic metaphase chromosomes permitted the separation of meadow fescue chromosomes based on their size (relative 4,6-diamino-2-phenylindole [DAPI] fluorescence intensity). The histogram of the chromosome DAPI fluorescence (flow karyotype) of meadow fescue ($n = 7$) consisted of three distinct peaks (Fig. 1). All seven chromosomes of meadow fescue can be individually identified based on their specific patterns of hybridization with various DNA probes (Kopecký et al., 2010). This permitted the identification of particles sorted from individual peaks on flow

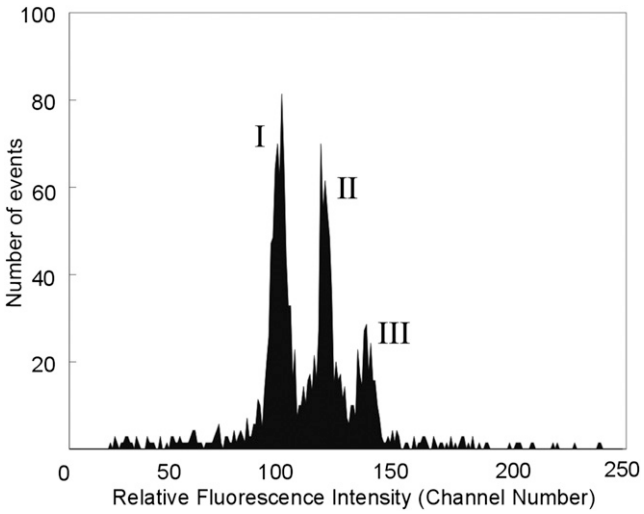


Figure 1. Flow karyotype. The distribution of relative chromosome fluorescence intensity (flow karyotype) was obtained after analysis of a DAPI-stained suspension of mitotic chromosomes of meadow fescue ($2n = 2x = 14$). The flow karyotype consists of two composite peaks, I and II, representing chromosomes 1F, 5F, and 6F and chromosomes 2F, 3F, and 7F, respectively, and peak III, representing chromosome 4F. This chromosome can be discriminated and sorted individually.

karyotype. Thus, chromosomes 1, 5, and 6 were sorted as a group from peak I, chromosomes 2, 3, and 7 formed peak II, and chromosome 4 formed peak III. On average, we were able to collect about 8×10^3 copies of chromosome 4F from a sample prepared from approximately 150 root tips. For DNA amplification and subsequent sequencing, we collected about 40,000 4F chromosomes. The purity of isolated chromosomes was estimated by FISH. Based on screening 300 sorted chromosomes from two randomly chosen samples, the average purity was 92.4%, with contamination by almost all remaining chromosomes in low frequencies (chromosome 1, 0.36%; chromosome 2, 2.18%; chromosome 3, 2.54%; chromosome 5, 0.36%; and chromosome 7, 2.18%).

Amplification and Sequencing of Chromosome 4F

An equivalent of 50 ng of DNA was obtained by flow cytometric sorting of 40,000 copies of chromosome 4F. This yielded 25 ng of purified DNA, which was used in three independent multiple displacement amplification (MDA) reactions, providing a total of 12.1 μg (4.1 μg + 4.2 μg + 3.8 μg) of DNA. For sequencing, individual samples of amplified DNA were combined to reduce a possible bias introduced by the MDA. A total of 4 μg of MDA DNA from chromosome 4F was used for Illumina HiSEquation 2000 sequencing. A total of 85,351,865 paired-end sequence reads with lengths of 101 bp were generated, yielding 28,553 Mb of sequence. Considering the size of chromosome 4F at 543 Mb (Kopecký et al., 2010), this represented approximately 50× coverage.

The assembly of the Illumina paired-end reads was performed using SOAPdenovo (Luo et al., 2012) with different k-mer sizes (i.e. 21–83 k-mer). The assembly with the best L50 (a weighted median statistic such that 50% of the entire assembly is contained in contigs equal to or larger than this value; 70-mer, 132 bp) and a maximum contig length of 15,919 bp was chosen for further analyses. The low L50 value, the large number of short contigs, and the high amount of repetitive elements indicate a highly fragmented assembly. Removing the repetitive sequences from the assembly, the L50 value increases considerably (1,623 bp). This suggests that the nonrepetitive genome space is well assembled and provides a useful resource for chromosome structure analysis, gene detection, and construction of a virtual ordered gene map.

Gene Content of Chromosome 4F

To estimate the number of genes on 4F, sequence comparisons against the genomes of *B. distachyon*, rice, and sorghum were made. We used stringent BLASTX searches that required 75% or greater (*B. distachyon*)/70% (rice, sorghum) sequence similarity over at least 30 amino acids (Table I). Between 2,629 and 3,056 significant matches were found. A total count of cumulative, nonredundant matches on 4F was 4,626. Considering the estimated molecular length of 4F at 543 Mb and the gene density on 4F representative for the entire meadow fescue genome (Kopecký et al., 2010), this would scale up to 27,048 genes. It should be mentioned that this number does not consider the putatively nonhomologous genes.

Table I. *GenomeZipper results*

General overview of the GenomeZipper results of the 4F chromosome. All numbers given are nonredundant.

Parameter	Chromosome 4F
No. of anchored gene loci	4,672
Lolium spp. SNP markers with match to syntenic genes	109 (out of 154)
No. of anchored barley full-length complementary DNAs	1,697
No. of anchored 4F contigs via first best hit	20,613
No. of anchored 4F contigs via bidirectional BLAST hit	3,700
No. of anchored *Festuca* spp. ESTs	1,444
No. of anchored *B. distachyon* genes	3,056
No. of anchored rice genes	2,629
No. of anchored sorghum genes	2,826

A Virtual Gene Map of Chromosome 4F

Because of the lack of suitable genetic markers for meadow fescue, 154 genetic markers from chromosome 4 of *L. perenne* were used as a backbone to construct a virtual gene map of chromosome 4F. *L. perenne* is a closely related species that evolved from the genus *Festuca* about 2.2 million years ago (Polok, 2007). Out of the 154 *L. perenne* markers, 109 (70.8%) could be associated to genes with conserved synteny under these conditions: (1) the best first hit, with (2) a minimal alignment length of 30 amino acids, and (3) an identity of 75% (*B. distachyon*)/70% (rice, sorghum). A virtual ordered gene map was developed using the GenomeZipper protocol (Mayer et al., 2011). With this approach, we identified 4,672 loci, of which 4,626 correspond to regions with conserved synteny in *B. distachyon*, rice, and sorghum (Table I). Among these, 1,515 (32.7%) loci were supported by genes in all three reference organisms, 855 (18.5%) in two of them, and 2,256 (48.8%) genetic loci were supported by only one gene with conserved synteny (Fig. 2). The higher number of syntenic *B. distachyon* genes reflects a closer phylogenetic relationship of meadow fescue to *B. distachyon* than to rice and sorghum. Furthermore, 1,697 barley full-length complementary DNAs were uniquely associated to either a marker sequence or at least one syntenic gene (Table I).

Comparative Analysis of Chromosome 4F with Relatives

Using GenomeZipper, we performed a detailed comparative analysis of chromosome 4F with sequenced genomes of *B. distachyon*, rice, sorghum, and barley. An ancient translocation differentiates chromosome 4F from 4H of barley (Fig. 3). Chromosome 4F is collinear

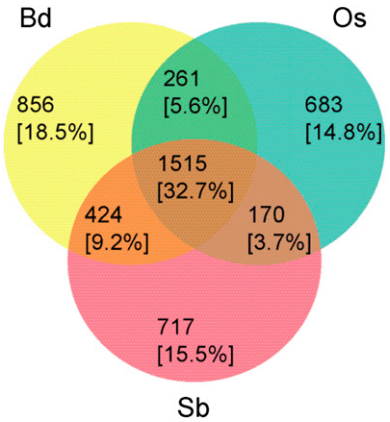


Figure 2. Model species genes anchored to 4F. The Venn diagram shows the number of *B. distachyon* (Bd), rice (Os), and sorghum (Sb) genes that are anchored in the virtual gene map of meadow fescue chromosome 4F. Overlaps of the circles show the number of genes anchored at the same locus. [See online article for color version of this figure.]

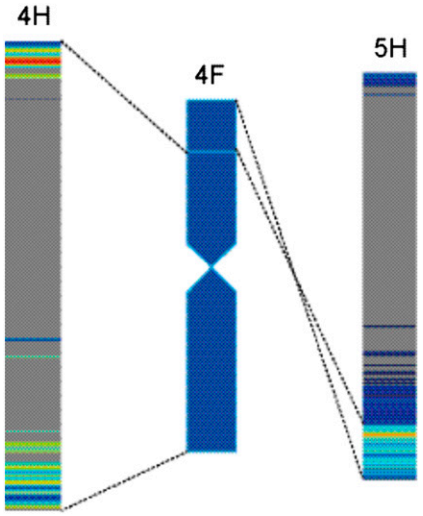


Figure 3. Collinearity of 4F with the barley genome. Meadow fescue 4F Illumina contigs were mapped on the physical map of barley chromosomes 4H and 5H. Syntenic regions are colored yellow-red, whereas nonsyntenic regions are colored dark blue. This approach clearly identifies the collinearity of chromosome 4F with barley chromosome 4H and the terminal part of 5HL. The connectors used indicate the orientations of the particular parts.

with the entire chromosome 4H but also with the terminal region of the long arm of chromosome 5H (5HL). Based on the position of genetic markers, the 5H syntenic segment is in an inverted orientation. For sequences generated from 4F, we identified in silico syntenic regions on *B. distachyon* chromosomes 1 and 4 (Fig. 4), rice chromosomes 3 and 11 (Supplemental Fig. S1), and sorghum chromosomes 1 and 5 (Supplemental Fig. S2).

Comparative mapping results between physical maps of barley chromosomes 4H and 5H, *Festuca* chromosome 4F, *Lolium* chromosome 4L, and wheat chromosome 4A show that the terminal segment originally from the ancestral chromosome 4 and now located at 5HL is not identical to the segment involved in the 4AL/5AL translocation (Naranjo et al., 1987; Supplemental Fig. S3). Our observation is similar to that of Alm et al. (2003). The location of the corresponding segment from 5HL is found on short arms of *Festuca* spp. 4F and *Lolium* spp. 4L chromosomes (both carrying ancestral type chromosome 4), while in wheat, the long arm of chromosome 4A is involved (Supplemental Fig. S3). The 5H regions syntenic with 4F and 4A are from 127.96 and 159.79 cM to the telomere (196.85 cM), respectively. While *Lolium* spp. 4L and *Festuca* spp. 4F appear structurally alike, a comparison of *Festuca* spp. 4F against wheat 4A indicates numerous inversions and translocations (Fig. 5). The extensive collinearity between *Lolium* and *Festuca* spp. along with the known genome structure of *Lolium* spp. (Pfeifer et al., 2013) suggest that the collinear 5HL segment on *Festuca* spp. chromosome 4 represents a

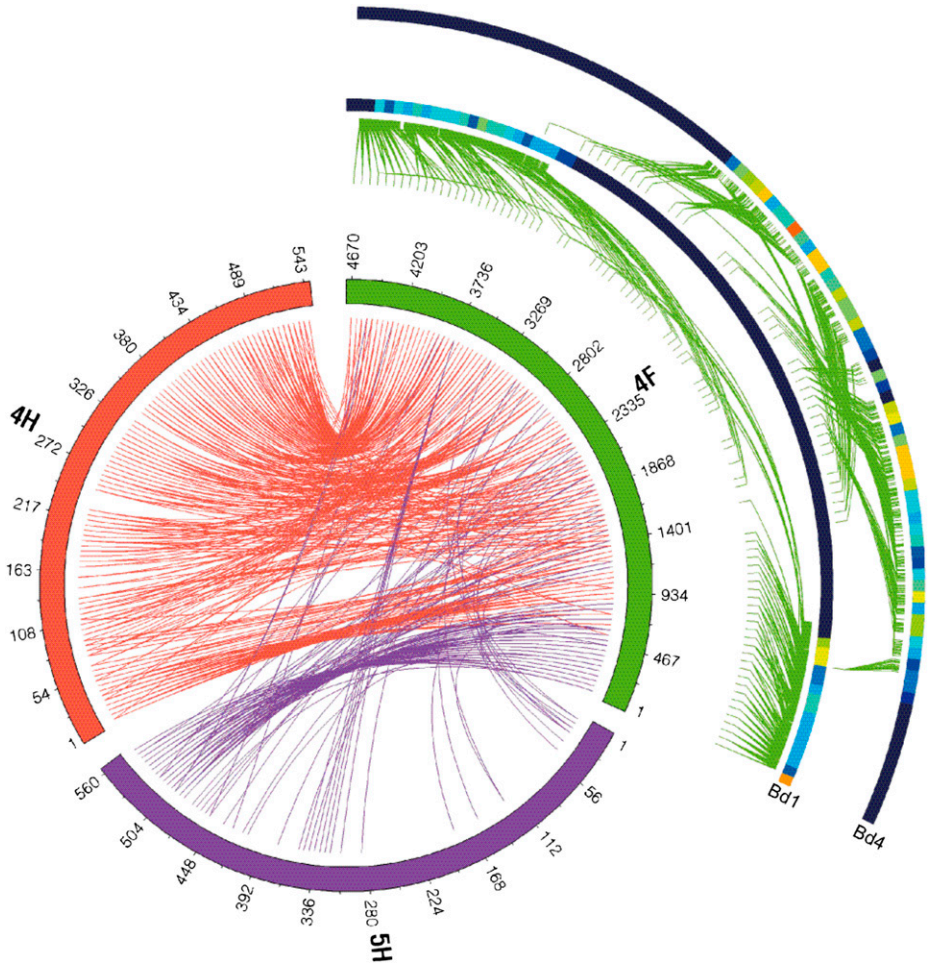


Figure 4. Collinearity of 4F with *B. distachyon*. High-density comparative analyses show the linear gene order of the meadow fescue GenomeZipper versus the sequenced genomes of barley and *B. distachyon*. The concentric circles are as follows. The inner circle represents chromosome 4F (green) and collinear chromosomes 4H (red) and 5H (purple) of barley. The outer circles represent homologous chromosomes of *B. distachyon* (Bd1 and Bd4). The heat maps illustrate the density of genes hit by the contigs from the 4F chromosome. Syntenic regions are colored red, whereas nonsyntenic regions are colored blue. Putative orthologs between 4F and barley chromosomes 4H and 5H are connected with lines. The scale is given in gene loci for chromosome 4F and in megabase pairs for barley chromosomes 4H and 5H.

unidirectional translocation in barley rather than a reciprocal translocation. In the case of reciprocal translocation, we would expect to observe some sequences being present in the terminal region of the short arm of barley chromosome 4H (4HS) and absent in 4F (Figs. 3 and 5A). However, this was not the case, and all sequences detected in the terminal region of 4HS were also present on chromosome 4F. This interpretation is consistent with observations made in *Lolium* spp. (Pfeifer et al., 2013). Since *Festuca* (and *Lolium*) spp. chromosome 4 is collinear with rice chromosome 3, which indicates more ancestral chromosomal structure, we hypothesize that this unidirectional translocation occurred in the Triticeae branch after the *Festuca/Lolium* genera split.

The linear ordered gene maps of *Festuca* spp. chromosome 4F, wheat chromosome 4A, and barley chromosome 4H allow one to analyze the number of syntenically conserved genes against the reference genomes of *B. distachyon*, rice, and sorghum (Fig. 6). The proportion of shared syntenic genes on all three chromosomes ranges between 23% and 28%, while between 19% and 24% of the genes are found for two of the chromosomes and between 49% and 57% are found for only one of the chromosomes analyzed. Strikingly higher percentages of genes conserved in *Festuca* spp. 4F but not for barley 4H and wheat 4A (30%–47%) were found illustrating the higher degree of syntenic conservation and, at least in part, a closer evolutionary relationship of *Festuca* spp. against the reference genomes used.

Repetitive DNA of Chromosome 4F

With the aim to determine the major repetitive DNA constituents of chromosome 4F, repeat reconstruction was done on Illumina data representing 1× coverage of chromosome 4F (550 Mb). Graph-based clustering resulted in 252,144 clusters containing 4,224,228 sequence reads (76%). A total of 855 clusters contained at least 100 sequences, from which 450 clusters contained more than 555 sequences (0.01% of analyzed reads; Supplemental Fig. S4). The largest clusters were manually annotated using a combination of multiple approaches based on similarity searches against sequence databases and graph layout analysis. The aim was to identify tandem organized repeats useful as

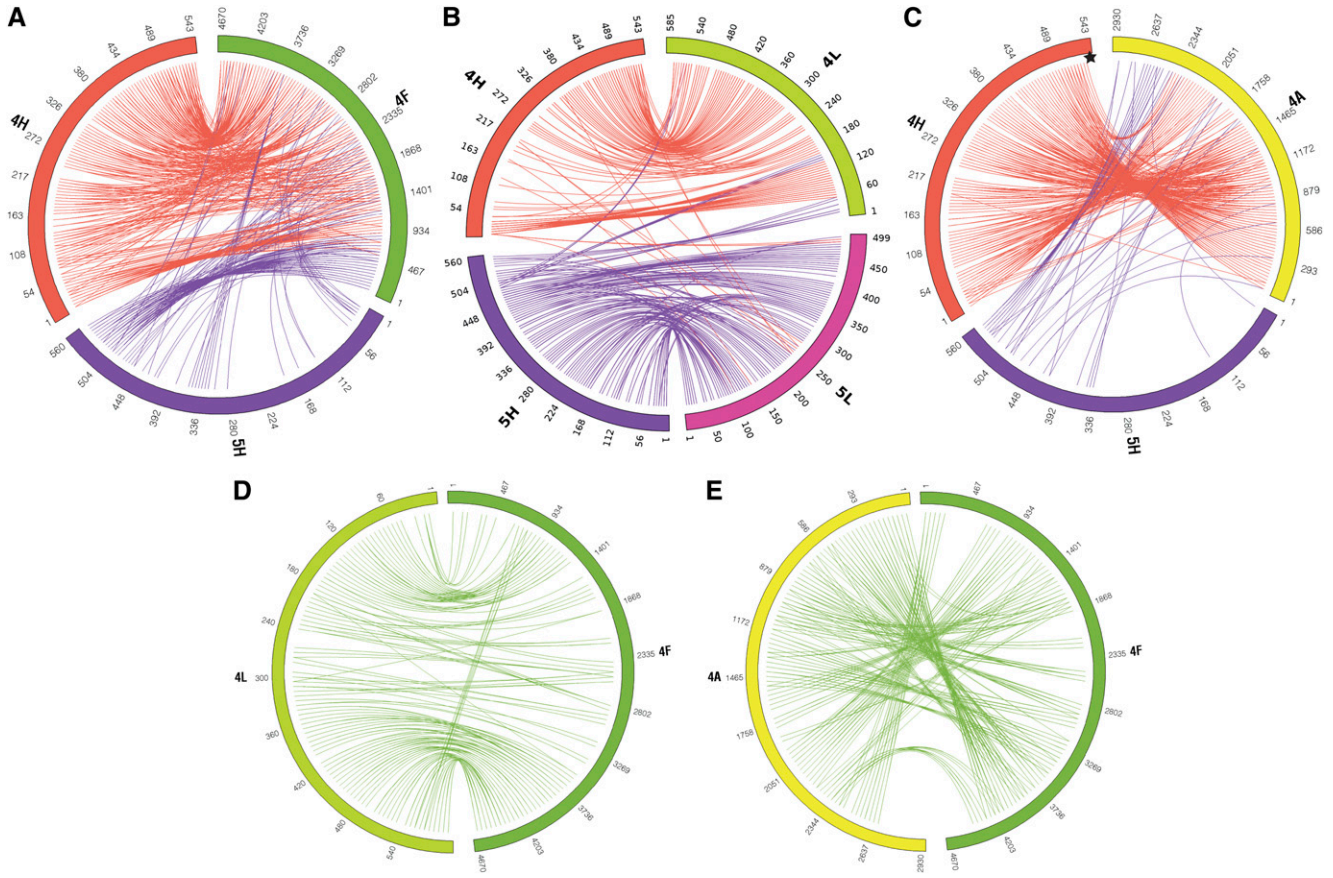


Figure 5. Chromosome 4/5 translocation. The top three circles (A–C) show the chromosome 4/5 translocation in three different grass chromosomes (*Festuca* spp. 4F, wheat 4A, and *Lolium* spp. 4L) in comparison with the corresponding homologous chromosomes 4 and 5 in barley. The location of the segment from 5HL for both *Festuca* (A) and *Lolium* (B) spp. is found on the short chromosome arm, while in wheat (C), the long arm of chromosome 4A is involved. The observed chromosome 4/5 translocation is a unidirectional translocation in *Festuca* and *Lolium* spp. rather than a reciprocal translocation, as found in wheat (Alm et al., 2003). The bottom two circles depict the collinearity between meadow fescue chromosome 4F and *L. perenne* chromosome 4L (D) and wheat chromosome 4A (E). While *Lolium* and *Festuca* spp. are highly collinear, the synteny between wheat and *Festuca* spp. is interrupted by several inversions and translocations.

species-specific and/or chromosome-specific cytogenetic markers.

Annotation of the largest clusters led to the classification of approximately 53% of the 4F chromosome sequence. The largest clusters contained mainly Ty3/Gypsy-like elements, which were found to be the most abundant repeats (28.64% of analyzed sequence reads) of chromosome 4F. Ty1/Copia-like elements represented less than 7% and DNA transposons represented about 3% of annotated sequences (Fig. 7). Interestingly, we were not able to characterize sequences within several large clusters that account for more than 8% of the analyzed sequence reads of chromosome 4F. These sequences can represent unknown DNA repeats and/or whole-genome amplification artifacts (Fig. 7).

Based on the graph layout (Fig. 8), contigs derived from clusters containing putative tandem repeats were identified and further investigated using similarity dot-plot analysis (Supplemental Fig. S5). The analysis revealed the presence of tandem or semitandem organized repeats in 43 clusters, which together represented about 2.71% of the chromosome 4F sequence.

Experimental Investigation of Tandem Organized Repeats

Out of 43 in silico-identified tandem repeats, sequences of 15 putative tandem repeats containing repetitive units of different length were investigated in more detail to study their organization in the *Festuca* and *Lolium* spp. genomes (Supplemental Table S1). A set of primers specific for tandem units was designed and used for the preparation of probes for FISH and for Southern hybridization. PCR products obtained after amplification on genomic DNA of meadow fescue as the template DNA were sequenced using dideoxy-chain termination reaction to check their authenticity. Moreover, PCR was done on flow-sorted chromosomes of

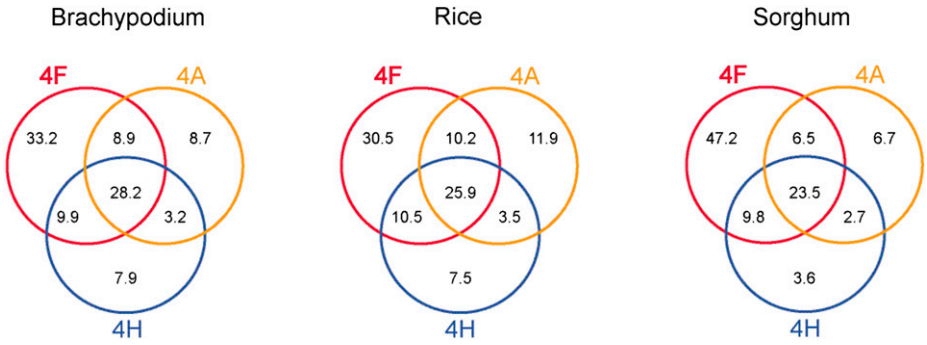


Figure 6. Conserved syntenic regions in the virtual linear ordered maps of *Festuca* spp., barley, wheat, and *Lolium* spp. chromosome 4 as defined by comparison with model grass genomes. The intersecting circles of the Venn diagrams depict the percentage of syntenic conserved reference genes (*B. distachyon*, rice, sorghum) shared by meadow fescue chromosome 4F (red), wheat chromosome 4A (orange), and barley chromosome 4H (blue). [See online article for color version of this figure.]

meadow fescue and on *L. perenne* genomic DNA to verify species and/or chromosome specificity. The results showed that all tandem repeats were present in all chromosome peaks of meadow fescue flow karyotype and that positive PCR products were obtained for all 15 selected tandem repeats also on *Lolium* spp. genomic DNA (Supplemental Table S1).

To confirm the tandem character of the repeats within the clusters, Southern hybridization was used to study the organization of repeats from the clusters that were successfully mapped on *Festuca* spp. mitotic chromosomes using FISH (see below). A ladder-like pattern that is typical for tandem organized repetitive units was obtained for all eight putative tandem repeats that gave visible signals after FISH (see below; Fig. 9). As shown in Figure 10, the tandem repeats identified in sequences of chromosome 4F of meadow fescue are also present in the nuclear genome of *L. perenne*, although at lower copy number.

Physical Localization of Repeats Using FISH

Out of 15 tandem repeats tested (fpTR1–fpTR15), we were able to physically map five fpTRs on chromosome 4F using FISH (Fig. 9; Supplemental Table S1). All of them produced hybridization patterns also on other chromosomes. Additionally, the other three fpTRs mapped on chromosomes other than 4F (fpTR6, fpTR11, and fpTR12; Supplemental Table S1). This might be caused by the sensitivity of FISH. None of the fpTRs produced signal

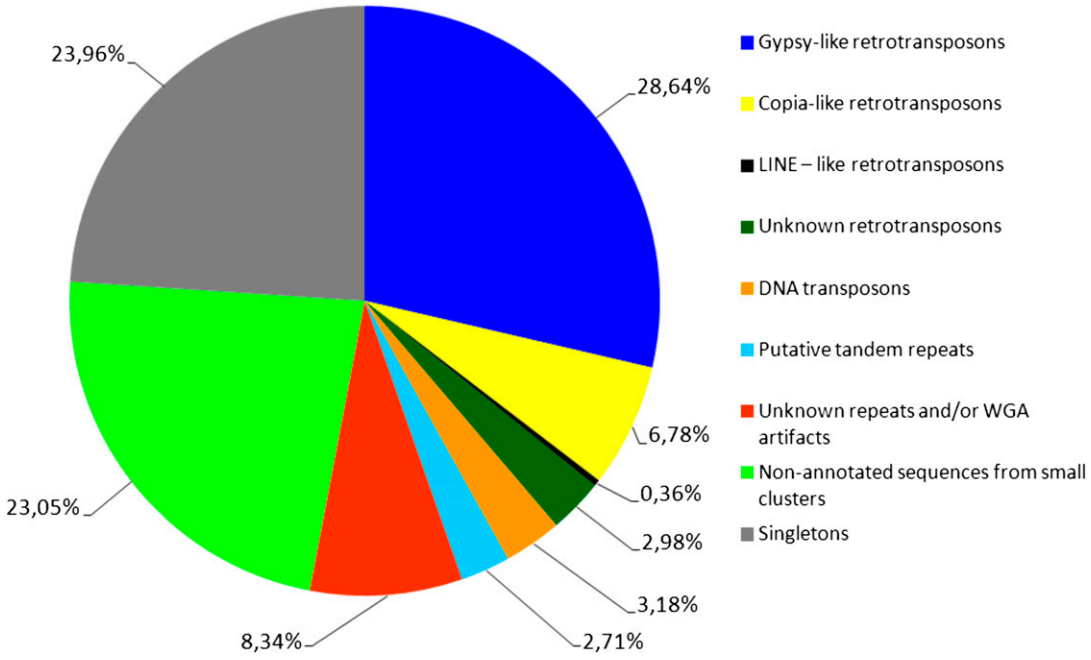


Figure 7. Repeats on 4F. The proportion of different repetitive elements in sequence reads of 4F chromosome is shown. [See online article for color version of this figure.]

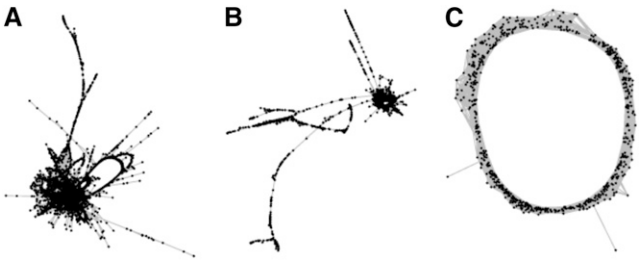


Figure 8. Graph layouts visualized by SeqGrapheR. Examples of graph layouts were calculated using the Fruhterman-Reingold algorithm for different types of tandem organized repeats: fpTR1 (A), fpTR5 (B), and fpTR15 (C). Dots and lines represent sequence reads and similarity hits between them, respectively.

exclusively on 4F; thus, they cannot be considered as chromosome specific. On the other hand, our observation opens the way for precise and robust molecular karyotyping of this species. As most of the fpTRs produced visible signals also on mitotic chromosomes of *L. perenne* and F1 hybrid *L. perenne* × meadow fescue (Fig. 9), the repeats may find even broader use as probes for FISH in cytogenetics of the *Festuca-Lolium* spp. complex.

Using FISH, we were able to localize a telomeric repeat on *Festuca* spp. chromosomes. Surprisingly, it produced signals not only in the telomeric regions of both chromosome arms of 4F but also interstitially (Fig. 9). Such interstitial signals were also detected on chromosome arms 2FL, 5FL, and 6FS. In general, FISH done on sorted chromosomes displayed higher sensitivity and produced more bands with higher resolution on individual chromosomes than on standard squashed preparations (Fig. 9).

DISCUSSION

Despite their considerable economic importance, forage and turf grasses lag far behind many other crops in terms of genetics, genomics, and bioinformatics. The delay is due to many factors, but outcrossing character, population-based breeding, and frequent aneuploidy do not help. Moreover, genomes of species within *Festuca-Lolium* are large and complex. The genome of meadow fescue was estimated at 1C = 3,175 Mb by DNA flow cytometry and is similar in size to that of human (Doležel et al., 2003; Kopecký et al., 2010). The ryegrass genome is just a little smaller (1C = 2,623 Mb for *L. perenne* and 1C = 2,567 Mb for *L. multiflorum*). Because of the genome complexity, any approach reducing sample complexity is welcome in genomic studies.

Partitioning the Grass Genomes

Analysis of a complex genome where a majority of DNA is represented by repeats can be simplified by several approaches. There are methods available that avoid sequencing of the repetitive parts of genomes. Of these, sequencing of complementary DNA to generate ESTs is among the most successful. However, ESTs fail to sample rare or conditional transcripts (Martienssen et al., 2004), and other methods were proposed to target the gene space in large and complex genomes, such as Cot fractionation and methyl filtration (Rabinowicz et al., 1999; Peterson et al., 2002). Unfortunately, the latter two methods did not meet the expectations.

Another alternative to tackle the complex genomes is to dissect them into smaller elements and sequence these elements individually. Working with naturally uniform and independent units, such as chromosomes, is perhaps the most powerful approach. In hexaploid wheat, individual chromosomes represent only 3.6% to 5.9% of the entire genome (Doležel et al., 2009), and even in a diploid species such as meadow fescue, isolation of individual chromosomes dissects the 3.2-Gb genome into 373- to 543-Mb units, each representing 11.7% to 17.1% of the entire genome (Kopecký et al., 2010).

Individual chromosomes can be isolated in two ways: by microdissection and flow sorting. Microdissection gives access to any chromosome or a chromosome segment. However, this is a very tedious approach, and for all practical purposes, the total yield is limited to only a few copies of a particular chromosome (Zhou and Hu, 2007). On the other hand, flow cytometry can isolate high copy numbers of the same chromosome with purity usually exceeding 90%. Flow sorting relies on differences in chromosome size or, rather, on the difference in relative fluorescence intensity. The output of flow cytometric analysis is a histogram of relative chromosome fluorescence intensity (reflecting chromosome size), which is called a flow karyotype (Fig. 1). Ideally, each chromosome is represented by a single peak on the flow karyotype. However, similarities in chromosome size within a genome usually result in the appearance of composite peaks representing two or more chromosomes. According to Doležel et al. (2009), there has to be at least a 10% difference in chromosome size to generate a separate peak on a flow karyotype. Unfortunately, in most plant species, chromosomes are not that different in size. In hexaploid wheat (21 chromosome pairs), the flow karyotype consists of only four peaks, and only one of those contains a single chromosome, 3B (Vrána et al., 2000). Similarly, the genome of meadow fescue was dissected here into three peaks, where only one peak represented a single chromosome type (chromosome 4F). However, the plasticity of plant genomes (especially in polyploids) makes it possible to develop special cytogenetic stocks with reconstructed karyotypes. Single chromosome substitution lines of *L. multiflorum*-meadow fescue, developed in our previous work, may provide an option for sorting other chromosomes of meadow fescue due to a sufficient difference in chromosome length (Kopecký et al., 2010).

Virtual Gene Order on Chromosome 4F

The GenomeZipper approach applied to individual chromosome sequences is a powerful tool for comparative studies. It significantly increases the resolution level relative to genetic markers. In the previous report

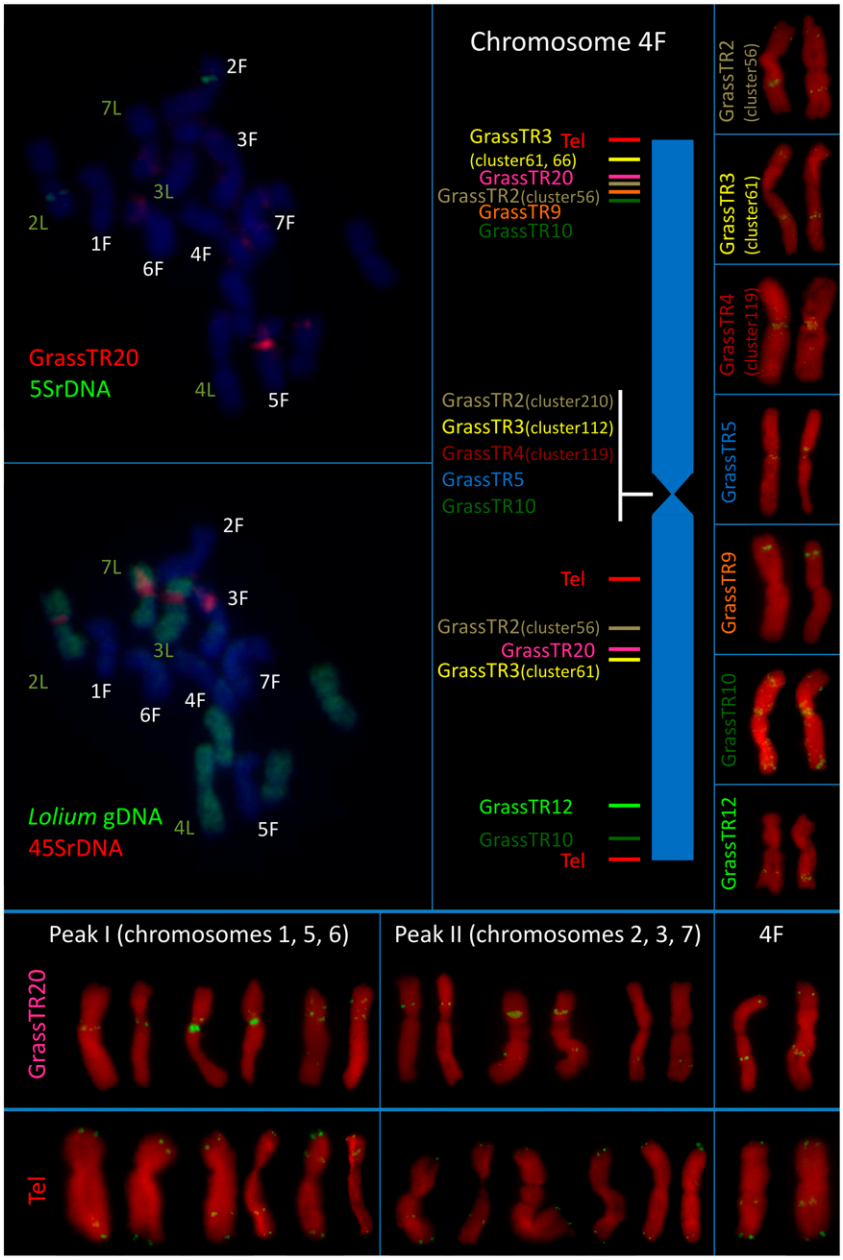


Figure 9. Cytogenetic mapping of repeats. Cytogenetic mapping shows newly identified tandem repeats and telomeric repeats on sorted chromosomes of meadow fescue and the F1 hybrid of *L. perenne* × meadow fescue.

on the collinearity of the meadow fescue genome with rice, sorghum, *Lolium* spp., oat (*Avena sativa*), maize (*Zea mays*), and Triticeae, a limited number of markers were used (Alm et al., 2003). Prior to this study, only 36 markers were available from chromosome 4F, with only a subset (seven to 30) suitable for any pairwise comparison. This contrasts with thousands of sequences from individual chromosomes (4F here) of model plant species (rice, sorghum, *B. distachyon*, and barley).

Estimation of Gene Content

The GenomeZipper approach permitted the estimation of gene content in meadow fescue and, by implication, all related *Festuca* spp. Assuming the estimated size of chromosome 4F at 543 Mb and the 4F gene density representative for the entire genus *Festuca* genome (Kopecký et al., 2010), the total genus *Festuca* genome contains about 35,000 genes. This is a slightly higher number that the estimation for barley (32,000 or more genes; Mayer et al., 2011) and less than that for the B genome of wheat (38,000 genes; Choulet et al., 2010). The recent estimate for the A genome of wheat is below that for *Festuca* spp. and barley (28,000 genes; Hernandez et al., 2012). In *L. perenne*, the most closely related species for which DNA sequence data are available, almost 25,700 matches with *B. distachyon* genes were obtained (Byrne et al., 2011).

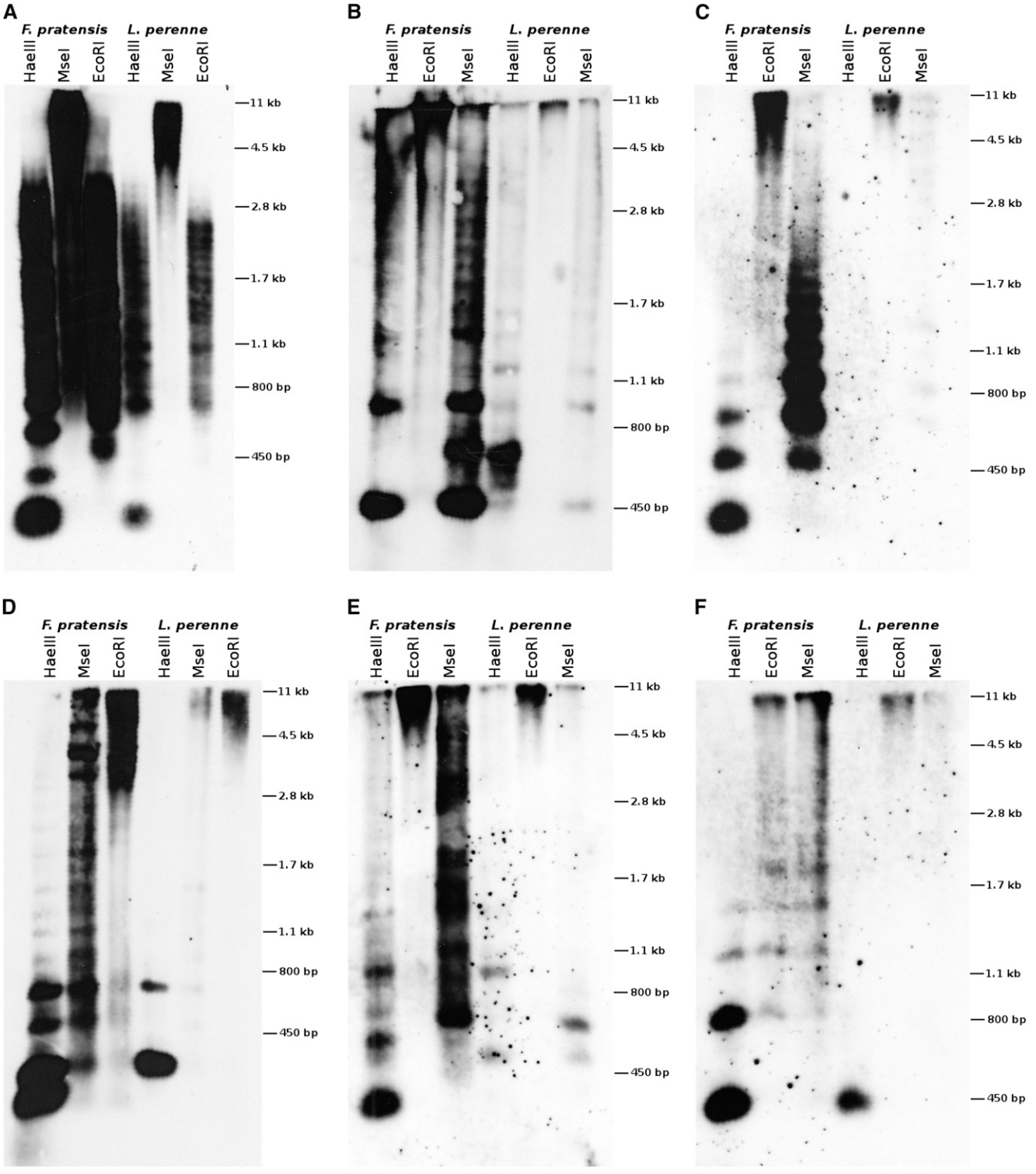


Figure 10. Southern hybridization with the probes for tandem repeats. Examples show Southern hybridization of meadow fescue and *L. perenne* genomic DNA digested using three different restriction endonucleases with the probes for tandem organized repeats fpTR1 (A), fpTR4 (B), fpTR5 (C), fpTR7 (D), fpTR12 (E), and fpTR15 (F). The ladder-like pattern indicates the presence of tandem organized repetitive units and supports the results of dotter analysis as well as the FISH analysis.

Apart from estimating gene content, the annotated sequences represent an invaluable genome resource for molecular biologists and breeders of fescue. For example, the sequences can be screened for the presence of resistance gene analogs. Because of a general lack of molecular markers for fescue, this new resource opens avenues for in silico identification of sequences suitable as DNA markers, including simple sequence repeat, presence/absence variation, and single-nucleotide polymorphism (SNP), and help in developing SNP platforms for marker-assisted breeding of this agriculturally valuable species.

4F Is an Ancestral Chromosome toward Modern Cereals

Based on the comparative analysis, chromosome 4F appears collinear with the entire barley chromosome 4H as well as with the terminal region of 5HL (Fig. 3). The observed homology of 4F to chromosome arms 4S, 4L, and 5L from Triticeae is consistent with previous observations by Alm et al. (2003) based on mapping of molecular markers. Chromosome 4F is collinear with the entire chromosome 3Os of rice (including the centromeric introgression of chromosome 11Os). On the other hand, 3Os is collinear with chromosome 4H and also with the terminal region of 5HL (Gale and Devos, 1998). Thus, meadow fescue chromosome 4F is likely more ancient than barley chromosome 4H. In cereal genomes (the A, B, and D genomes of wheat and the H genome of barley), the direct ancestral precursor that is syntenic with rice chromosome 3Os was broken and the terminal part of the short arm was translocated to the distal part of the long arm of chromosome 5, resulting in the present-day chromosome 5H. This happened after the divergence of the *Festuca* and *Lolium* genera from the grass lineage between 25 million years ago (divergence of Triticeae and Poeae) and 11.6 million years ago (divergence of barley and wheat genomes; Chalupska et al., 2008). In *Festuca* and *Lolium* spp., the direct ancestral precursor syntenic with 3Os of rice remained intact and resulted in the present-day 4F and 4L chromosomes of *Festuca* and *Lolium* spp., respectively, as originally proposed by Alm et al. (2003). Therefore, chromosome 4F, in its current form, can be assumed as a potential transient chromosome 4H′ proposed by Thiel et al. (2009). Interestingly, a 4-fold increase in recombination frequency was observed in the region surrounding the translocation break point (Kopecký et al., 2010). This may indicate an unstable region of the chromosome more prone to chromosome breakage.

Newly Identified Repeats Are a Valuable Source of Cytogenetic Markers

Cytogenetic mapping involves the physical localization of a sequence of interest directly on chromosomes. Cytogenetic mapping has many applications, ranging from the study of structural chromosome changes (Mandáková and Lysák, 2008) to the determination of the positions and orientation of unassembled BAC contigs to support the development of physical maps (Pedrosa-Harand et al., 2009). In our specific case, cytogenetic markers capable of precise and unambiguous identification of chromosomes are helpful for purity checks of sorted chromosome fractions.

Various types of tandem repeats are valuable sources of useful cytogenetic markers. Satellites are tandemly repeated sequences with a repeat unit greater than 25 bp. Satellites of the *Afa* family have been used for identification of the D-genome chromosomes in wheat (Rayburn and Gill, 1986) and of barley chromosomes (Tsujimoto et al., 1997). Other satellites were used for karyotyping in barley (Brandes et al., 1995), *Avena* spp. (Katsiotis et al., 1997), banana (*Musa* spp.; Hřibová et al., 2007), and other plant species (Sharma and Raina, 2005) because they tend to form clusters, which facilitates their detection and positive chromosome identification. In this study, 11 newly characterized satellites provided localized signals on one or more chromosomes of meadow fescue and could serve as new cytogenetic markers in this species and probably also in ryegrasses.

Chromosomes of *Festuca* and *Lolium* spp. can be readily discriminated in hybrids using genomic in situ hybridization (Thomas et al., 1994; Kopecký et al., 2008a, 2008b). This method enables the visualization of parental chromatin in any stage of the mitotic cycle in natural allopolyploids or synthetic wide hybrids (Schwarzacher et al., 1989). It is assumed that this discrimination is based on the presence of abundant species-specific repetitive elements. From this perspective alone, it is of considerable interest to analyze the frequency and distribution of major repeats in parental species of a wide hybrid, to determine which of them, and in what proportions, contribute to the discrimination of parental genomes in hybrids. From observations using PCR made in this study, all newly identified tandem repeats were present in both genera, *Festuca* and *Lolium*. Southern hybridization and FISH revealed slight differences in the presence or signal intensity of individual fpTRs. Hence, none or just a few of the 15 tested appears responsible for distinguishing parental genomes. This indicates that tandem repeats are not responsible for the discrimination capacity of genomic in situ hybridization. However, newly identified tandem repeats provided new sources of cytogenetic markers and significantly increased levels of chromosome identification.

CONCLUSION

To our knowledge, this is the first report on the dissection of a complex and large forage grass genome using chromosome sorting. We sorted chromosome 4F of meadow fescue and sequenced it by Illumina. This provided a unique and rich resource to study the

genome organization of the species. The sequence data were used to estimate gene content, construct virtual gene order, and characterize repetitive elements. This approach provided an opportunity to describe synteny between chromosome 4F and genomes of model species (*B. distachyon*, rice, sorghum, and barley). Finally, we demonstrated the potential of sorted chromosomes for cytogenetic mapping of various repeats. Our results demonstrate that coupling the chromosome sorting and NGS technologies is a powerful approach that provides insights on chromosome structure and evolution at superior resolution.

MATERIALS AND METHODS

Plant Material

For chromosome sorting, seeds of meadow fescue (*Festuca pratensis* 'Fure'; $2n = 2x = 14$) were obtained from Dr. Arild Larson (Graminor). Seeds were germinated in the dark at 25°C distributed in petri dishes filled with distilled water for 5 to 7 d to achieve optimal root length (approximately 2–3 cm). For additional in situ hybridization experiments, *Lolium perenne* 'SR4220', meadow fescue cv Fure, and a diploid F1 hybrid of *L. perenne* cv SR4220 × meadow fescue cv Skawa were used. The plants was transferred to a hydroponic culture of Hydroponex at 0.9 g L^{-1} (Hu-Ben); after 5 to 7 d, actively growing root tips were collected to ice water for approximately 28 h, fixed in a 3:1 mixture of absolute ethanol and glacial acetic acid at 37°C for 7 d, stained in 1% acetocarmine for 2 h, and squashed in a drop of 45% acetic acid on clean microscope slides (Masoudi-Nejad et al., 2002).

Cell Cycle Synchronization and Accumulation of Metaphases

For chromosome flow sorting, synchronization of the cell cycle and accumulation of metaphases in root tips were performed according to Vrána et al. (2000). Distilled water in petri dishes was replaced by Hoagland nutrient solution (Gamborg and Wetter, 1975) with 1, 1.5, or 2 mM HU. After 18 h of incubation, the solution was replaced by a HU-free Hoagland solution. Samples of root tips were taken at 1-h intervals for up to 6 h, and the cell cycle synchrony and mitotic activity were examined with a microscope. To accumulate cells at metaphase, the seedlings were treated for 2 h with Hoagland solution containing 5 μM oryzalin after recovery from HU. Mitotic activity and metaphase frequency were analyzed on Feulgen-stained squash preparations.

Preparation of Chromosome Suspension

Chromosome suspension was prepared according to Doležel et al. (1992) with minor modifications. Seedlings were rinsed in deionized water and fixed in 2% (v/v) formaldehyde fixative made in Tris buffer supplemented with 0.1% (w/v) Triton X-100 at 5°C for 20 min. After washing in Tris buffer, about 150 root tips (approximately 2 mm long) were cut and transferred to a vial tube containing 750 μL of LB01 buffer (Doležel et al., 1989). The chromosomes were isolated after homogenization with a Polytron PT1300 homogenizer (Kinematica) at 20,000 rpm for 15 s. To remove large cellular fragments, the suspension was passed through a 50-μm pore size nylon mesh. Prior to flow cytometry, isolated chromosomes were stained with DAPI adjusted to a final concentration of 2 μg mL^{-1}.

Chromosome Analysis and Sorting

Chromosome analysis and sorting were done on a FACSAria II SORP flow cytometer (BD Biosciences) equipped with a 100-mW, 488-nm laser for scattered light detection and a 100-mW, 355-nm laser for DAPI excitation. Suspension of chromosomes was analyzed at rates of 800 to 1,000 events per second. Approximately 20,000 to 50,000 chromosomes were analyzed in each sample. DAPI fluorescence was collected through a 450/50-nm band-pass filter. To discriminate doublets, dot plots of DAPI fluorescence area versus DAPI fluorescence width were used. To ensure high purities of the sorted chromosomes, a tight gate was created around the chromosome 4F population, and chromosomes were sorted at rates of four to six chromosomes per second. To evaluate the purity of chromosome sorting, approximately 1,000 chromosomes were sorted onto microscope slides and tested by FISH. For DNA amplification, 40,000 chromosomes were collected into individual tubes and stored at −20°C until used.

Test of the Purity of Sorted Chromosomes Using FISH

To identify flow-sorted chromosomes and determine the extent of contamination of the sorted chromosome fractions, we performed FISH with probes for the 5S and 45S ribosomal DNA and two BAC clones (1G18 and 2N14). The two BAC clones were selected from a partial BAC library of meadow fescue (Kopecký et al., 2010). The DNA was labeled by the DIG-Nick Translation Kit or the biotin-Nick Translation Kit (Roche Applied Science) or by PCR labeling with biotin- or digoxigenin-labeled nucleotides (Roche). In situ hybridization was performed according to Kubaláková et al. (2003). Detection of hybridization sites was by anti-digoxigenin-fluorescein isothiocyanate (Roche) and streptavidin-Cy3 (Amersham), and counterstaining was with DAPI in the Vectashield antifade solution (Vector Laboratories). Preparations were screened with an Olympus AX70 microscope with epifluorescence and a SensiCam B/W camera.

Amplification and Sequencing of Chromosome 4F

Flow-sorted chromosomes were treated with proteinase, and their DNA was subsequently purified using Microcon YM-100 columns (Millipore) as described by Šimková et al. (2008) with minor modifications. Chromosomal DNA was amplified by MDA using the Illustra GenomiPhi V2 DNA Amplification Kit (GE Healthcare Bio-Sciences). Sequencing of amplified chromosomal DNA was performed with HiSEquation 2000 (Illumina). Four micrograms of MDA-amplified DNA was used to create the corresponding shotgun DNA-seq library. The library for the 4F chromosome was run in a single lane at Istituto di Genomica Applicata Technology Services.

Assembly of Chromosome 4F Sequences and Genome Zipper Analysis

The Illumina paired-end reads were assembled by SOAPdenovo (Luo et al., 2012) using different k-mer sizes. The result of the 70-mer run provided the assembly with the best sequence coverage and L50. Repetitive DNA content was identified using Vmatch (http://www.vmatch.de) against the MIPS-REdat Poaceae version 8.6.2 repeat library by applying the following parameters: 70% identity cutoff, 100-bp minimal length, seed length 14, exdrop 5, and e-value 0.001. The Munich Information Center for Protein Sequences repeat library contains known grass transposons from the Triticeae Repeat Database (http://wheat.pw.usda.gov/ITMI/Repeats) as well as de novo-detected long terminal repeat retrotransposon sequences from various grass species. After repeat masking and filtering out the repeats, 26.6% (1,081,325 sequences) of the sequences remained and were considered for the subsequent steps.

To assess the number of syntenic conserved genes present on meadow fescue chromosome 4, the repeat-filtered contigs were aligned against the protein sequences of *Brachypodium distachyon* (version 1.2), rice (*Oryza sativa*; rice RAP-DB genome build 4), and sorghum (*Sorghum bicolor*; version 1.4) by BLASTX. The following stringent filtering criteria were applied: (1) only the first best hit with (2) a minimal alignment length of 30 amino acids and (3) a minimal sequence identity of 75% (*B. distachyon*)/70% (rice, sorghum). A sliding-window approach (window size of 0.5 Mb, shift size of 0.1 Mb) was used to identify segments with conserved gene order, based on the density of homology matches between *Festuca* spp. and the reference genomes.

The extracted conserved genes from the three model grass genomes and the corresponding *Festuca* spp. contigs were structured and ordered to a virtual linear gene model by using the GenomeZipper approach (Mayer et al., 2011). The approach uses a genetic marker scaffold to compare and subsequently integrate and order syntenic conserved, homologous genes (orthologs) along the scaffold. Thereby, intervals defined by the genetic markers serve as anchor points to define the corresponding intervals from the reference genomes. Genes from the least distant reference genome (in this case, *B. distachyon*) get highest

priority for ordering, and subsequently, the additional reference genomes are overlaid onto this scaffold. Due to the absence of a high-quality genetic marker map of meadow fescue, genetic markers of *L. perenne* chromosome 4 were used as a scaffold to anchor the collinear segments from the model genomes (Studer et al., 2012).

To analyze chromosomal rearrangements between *Festuca* spp. and barley (*Hordeum vulgare*), chromosome 4F contigs were compared against the physical map of the barley genome (Mayer et al., 2012) using BLASTX. Filtering criteria were adjusted to the evolutionary distance between meadow fescue and barley (at least 85% identity and a minimum alignment length of 100 bp).

To test whether the observed *Festuca* spp. chromosome 4/5 translocation is structurally identical to the 4/5 translocation found in Triticeae (Naranjo et al., 1987), the *Festuca* spp. 4F contigs, ordered gene scaffolds for wheat (*Triticum aestivum*) 4A and *Lolium* spp. 4L, and barley chromosomes 4H and 5H were compared by bidirectional BLAST hits and visualized using Circos (Krzywinski et al., 2009).

To estimate the number of conserved syntenic genes shared by wheat 4A, barley 4H, and *Festuca* spp. 4F, the *B. distachyon*, rice, and sorghum homologs contained in the individual linear ordered gene maps were compared with each other. A Venn diagram to illustrate shared genes in all three chromosomes was generated for each reference genome.

Repeat Reconstruction and Annotation

A random data set representing 1× coverage of chromosome 4F (550 Mb) was extracted from the Illumina data and used for the reconstruction of repetitive elements using a graph-based method according to Novák et al. (2010) using the Louvain clustering algorithm (Blondel et al., 2008). Resulting clusters of sequence reads were manually annotated using several sources, including similarity searches with RepeatMasker, and BLASTX and BLASTN programs were used for similarity searches against public databases and also against the database of domains derived from plant mobile elements (http://repeatexplorer.umbr.cas.cz). Clusters represented as graphs were also analyzed using the SeqGraphR program (http://w3lamc.umbr.cas.cz/lamc/resources.php). Dotter (Sonnhammer and Durbin, 1995) was used to confirm the presence of tandem organized repeats.

Physical Localization of Repeats Using FISH

FISH on flow-sorted chromosomes was done according to Kubaláková et al. (2003), while the protocol of Masoudi-Nejad et al. (2002) was used for metaphase plates prepared from root tips of the F1 hybrid. Probes were prepared for 15 putative tandem repeats (fpTR1–fpTR15) identified in sequence data using PCR labeling with biotin- or digoxigenin-labeled nucleotides (Roche) and pairs of specific primers. For chromosome identification in metaphase of F1 hybrids, *L. perenne*, and meadow fescue, we used a 5S ribosomal DNA (rDNA) probe accompanied by reprobing of slides with a 45S rDNA probe and a probe made from genomic DNA of *L. perenne*. The latter two probes were made using the biotin-Nick Translation Kit and the DIG-Nick Translation Kit (Roche), respectively. For the 45S rDNA probe, DNA clone pTa71 (Gerlach and Bedbrook, 1979) containing a 9-kb *Eco*RI fragment of wheat rDNA, which carries the 18S-5.8S-26S cluster of ribosomal RNA genes, was used. The probe for 5S rDNA was prepared using PCR with a pair of specific primers (RICRGAC1 and RICRGAC2) that amplify 303 bp in rice (Fukui et al., 1994), using rice genomic DNA as a template. Additionally, a probe for telomeric repeats was prepared using PCR with $(AGGGTTT)_3$ and $(CCCTAAA)_5$ primers without a template. Reprobing was done as described by Schwarzacher and Heslop-Harrison (2000). Probe hybridization signals were detected by anti-digoxigenin-fluorescein isothiocyanate and avidin-Cy3 conjugates, and counterstaining was by 0.2 mg mL^{-1} DAPI in Vectashield antifade solution (Vector Laboratories). Slides were evaluated with an Olympus AX70 microscope equipped with epifluorescence and a SensiCam B/W camera. ScionImage and Adobe Photoshop software were used for processing of color images.

Southern Hybridization

Aliquots of genomic DNA samples corresponding to 3×10^8 of the nuclear genomes of meadow fescue cv Fure and *L. perenne* cv SR4220 were digested using restriction enzymes *Hea*III, *Mse*I, and *Eco*RI, size fractionated by 1.2% agarose gel electrophoresis, and transferred to Hybond N^+ nylon membranes (Amersham). Probes specific for selected putative tandem repeats were amplified using specific primers (Table I) and labeled with biotin. The Southern hybridization was done at 68°C overnight, and signals were detected using the BrightStar BioDetect kit according to the manufacturer's instructions to 90% stringency (Ambion), incubated with chemiluminescent substrate (CDP-Star; Amersham Biosciences), and exposed on x-ray film.

Sequence data from this article can be found in the GenBank/EMBL data libraries under accession numbers JX624129 to JX624136.

Supplemental Data

The following materials are available in the online version of this article.

Supplemental Figure S1. Collinearity of 4F with rice.

Supplemental Figure S2. Collinearity of 4F with sorghum.

Supplemental Figure S3. Collinearity of 4F, 4A of wheat, and 5H of barley.

Supplemental Figure S4. Distribution of chromosome 4F reads in sequence clusters.

Supplemental Figure S5. Dot plot of five tandem repeats.

Supplemental Table S1. Summary of the detection of tandem repeats.

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